Measuring an Equilibrium Constant

Chem 151 Lab

Solutions You Need From Last Week:

- 0.1 M Fe^{3+}
- 4 mM Fe^{3+}
- 6 mM Fe^{3+}
- 8 mM Fe^{3+}

- 0.2 mM SCN^{-}

Today's Equilibrium Reaction

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$

Initial	**Initial $[Fe^{3+}]$**	**Initial $[SCN^{-}]$**		**0**
Change	**- x**	**- x**		**+ x**
Equilibrium	**$[Fe^{3+}]_0$ - x**	**$[SCN^{-}]_0$ - x**		**x**

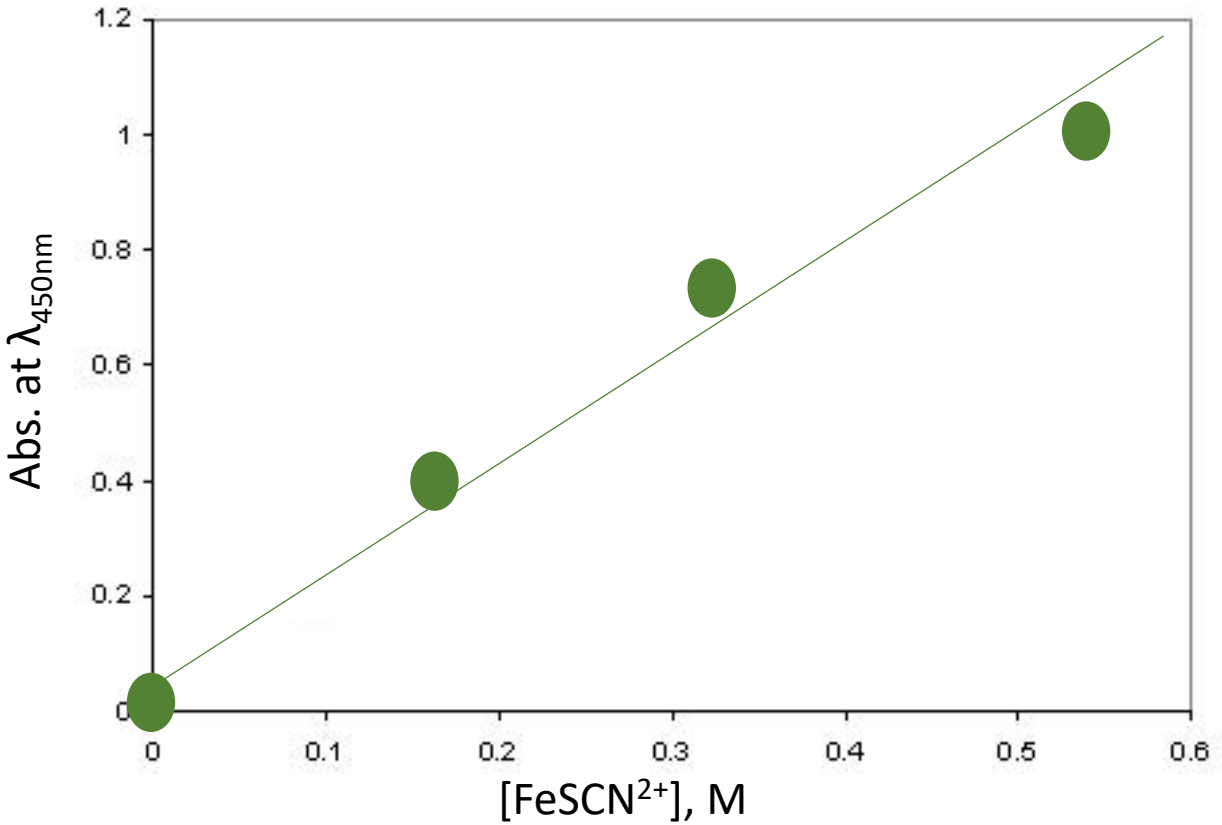
λ_{max} = 450 nm

Measured using Beer's Law

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

Part B: Calibration Standards

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$



Force Reaction

Then, $[FeSCN^{2+}]_{eq} = [SCN^{-}]_{initial}$

Dilution!

$M_1V_1 = M_2V_2$

Solve for M_2:

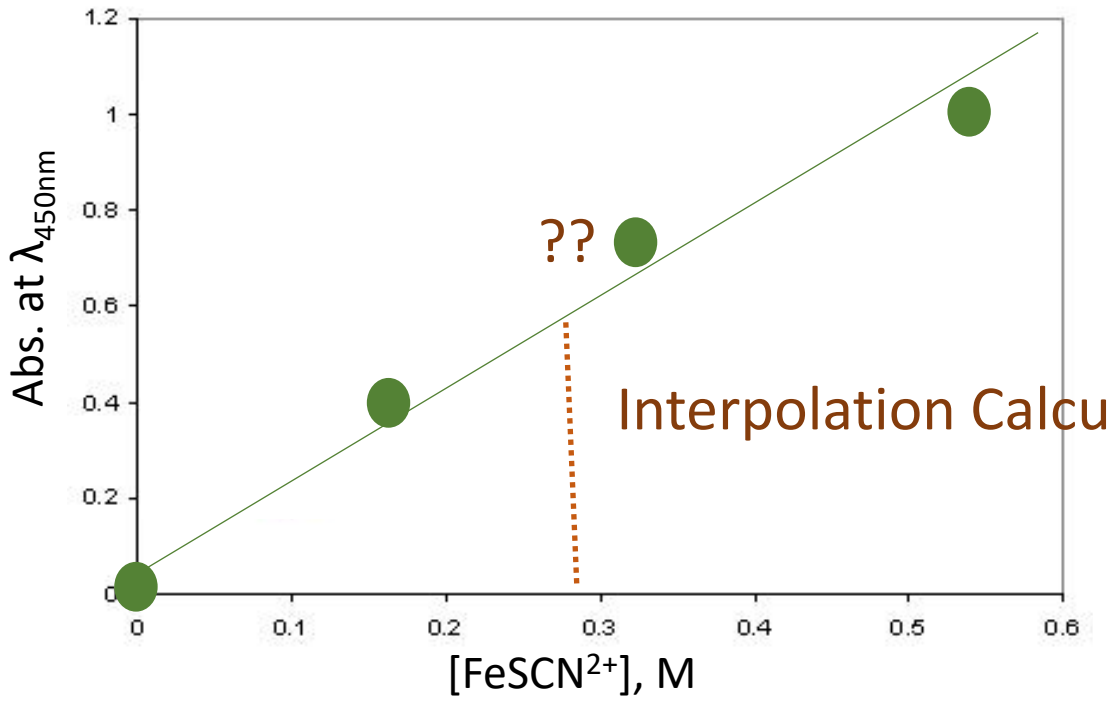
$$M_2 = \frac{M_1V_1}{V_2}$$

30 mL

*Use blank as $[FeSCN^{2+}] = 0$ as a data point!

Solns 1, 2, 3, + blank

Part C: Use the Calibration Plot from Part B



Measure soln 4, 5, 6

Interpolation Calculator = $[FeSCN^{2+}]_{eq}$ in each solution!

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$$

I	_____	_____		0
C	- x	- x		+ x
E	_____ - x	_____ - x		x

Need an ICE table for each soln!

You will calculate 3 K_c values.

$$K_c = \frac{[FeSCN^{2+}]_{eq}}{[Fe^{3+}]_{eq}[SCN^{-}]_{eq}}$$

Also find the average K_c

Logistics for Today's Lab

Have ALL solutions 1, 2, 3, 4, 5, 6 + blank ready and transferred in cuvettes before acquiring the spectrometer to collect data

*Make sure you dispose of all solutions into the proper waste containers in the hood

You can dispose of and clean all Nalgene bottle solutions